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 **Meta Review Status:** The meta review and decision have not been released yet.

 AI Review from DeepReviewer will be automatically processed

 Summary

This paper investigates the impact of using predictive models for need assessment in the German public employment service, a critical area where algorithmic decision-making is increasingly prevalent. The authors construct a synthetic dataset based on regional data from Bavaria in

2019, encompassing 85,299 individuals, to simulate the allocation of resources. Their methodology involves training predictive models, specifically logistic regression and XGBoost, to assess the need for aid and assistance. They then simulate resource allocation under three different scenarios: no intervention, a needs-based intervention where individuals with a predicted need probability of at least 50% receive resources, and a 'fair needs' intervention that ensures everyone has at least a 50% chance of receiving resources. The core finding of the study is that the use of predictive models, while intended to improve efficiency, can exacerbate existing social inequalities. Specifically, the authors observe that these models tend to deepen divides along the lines of migration background, education, and gender. This is demonstrated through an analysis of variance and income inequality metrics, which show significant disparities in resource allocation under the predictive model-based interventions. The paper highlights the ethical implications of using algorithms in public service provision, particularly concerning fairness and equity. The authors call for further research into the potential consequences of algorithmic decision-making, emphasizing the need to consider the broader societal impacts of these technologies. The study's significance lies in its empirical demonstration of how seemingly neutral algorithms can perpetuate and even amplify existing social inequalities, a crucial concern as these technologies become more widespread in public administration. The paper's contribution is timely and relevant, prompting critical reflection on the responsible use of AI in public services. However, the paper also suffers from several methodological and presentation issues that limit the strength of its conclusions. These include a lack of clarity in the definition of key variables, a lack of statistical rigor in the analysis, and a limited discussion of the study's limitations. These issues, which I will detail in the following sections, need to be addressed to strengthen the paper's contribution to the field.

✓ Strengths

The paper addresses a highly significant and timely issue: the ethical implications of using predictive models in public service provision, specifically within the German public employment service. This is a crucial area of inquiry as algorithmic decision-making becomes increasingly prevalent in public administration. The authors' focus on the potential for these models to exacerbate social inequalities is particularly important, as it highlights the need for careful consideration of fairness and equity in the design and implementation of such systems. The use of a real-world dataset from the German public employment service enhances the practical relevance of the study. By grounding their analysis in actual data, the authors provide a more realistic and compelling picture of the potential impacts of predictive models. This approach allows for a more nuanced understanding of the challenges and opportunities associated with the use of these technologies in a real-world context. The paper employs established machine learning models, namely logistic regression and XGBoost, which are widely used and well-understood. This choice of models ensures the technical rigor of the analysis and makes the findings more accessible to a broader audience. The use of standard evaluation metrics, such as accuracy, precision, recall, and AUC, further contributes to the technical soundness of the study. These metrics are commonly used in the field and provide a clear and objective way to assess the performance of the predictive models. The paper's exploration of different intervention strategies, including a 'fair needs' intervention, is also a strength. By comparing the outcomes of different allocation methods, the authors provide valuable insights into the potential for mitigating the negative consequences of predictive modeling. This comparative approach allows for a more nuanced understanding of the trade-offs associated with different resource allocation strategies. Overall, the paper's focus on a relevant and timely issue, its use of real-world data, its employment of established methods, and its exploration of different intervention strategies make it a valuable contribution to the field. However, as I will detail in the following sections, the paper also suffers from several weaknesses that need to be addressed to fully realize its potential.

✗ Weaknesses

While the paper addresses a critical issue, several weaknesses undermine its overall impact. First, the paper suffers from a lack of clarity and precision in its writing. The problem formulation is not clearly articulated, and the methodology and results are not explained in a way that is easy to follow. For example, the paper introduces terms like 'synthetic dataset' and 'predictive models' without sufficient context or definition, making it difficult for readers unfamiliar with these concepts to grasp the core ideas. The 'Method' section lacks a clear narrative flow, jumping between different aspects of the methodology without a logical connection. This lack of clarity makes it challenging to understand the authors' approach and the validity of their findings. Second, the paper fails to provide a clear definition of the outcome variables. While the 'Simulation'

section mentions 'labor market outcomes in the form of income, the likelihood of remaining unemployed at the end of the observation period, and the likelihood of claiming benefits and assistance,' these are not presented as formal definitions. The absence of a clear and precise definition of the outcome variables makes it difficult to interpret the results and understand their implications. The paper also uses undefined mathematical notation, such as ' Δ ' in Table 1, without explanation. This lack of clarity hinders the understanding of the results presented in the tables. Third, the methodology section lacks crucial details. The description of how resources are 'randomly assigned' is vague and lacks a precise mechanism. The paper states that 'Individuals with a probability of needs of at least 50% receive resources' under the 'Needs-based intervention,' but the exact process of random assignment is not detailed. Similarly, the 'Fair needs' intervention is described as ensuring 'everyone has at least a 50% chance to receive resources,' but the implementation of this random assignment is not explained. The baseline for comparison is also not explicitly stated, although it can be inferred to be the 'No Intervention' scenario. The choice of the 50% threshold is also not justified, further weakening the methodological rigor. Fourth, the results section lacks statistical rigor. The paper presents tables with mean values for metrics like variance and Gini coefficient, but there is no mention of confidence intervals, standard deviations, or statistical significance tests. The absence of these measures makes it difficult to assess the uncertainty associated with the results and whether the observed differences are statistically significant. This lack of statistical rigor weakens the conclusions drawn from the results. Fifth, the paper lacks a thorough discussion of its limitations. While the 'Limitations' section briefly mentions the regional nature of the data, it does not delve into potential biases in the synthetic data generation process or the generalizability of the findings to other contexts. The paper also fails to address how pre-existing societal inequalities might influence the results or the potential biases introduced during the synthetic data generation process. The sensitivity of the results to different model parameters or alternative synthetic data generation methods is also not explored. This lack of discussion limits the paper's overall impact and raises questions about the robustness of its findings. Finally, the paper's presentation relies heavily on tables, which can be difficult to interpret. The absence of visual aids, such as graphs and charts, makes the paper less accessible and engaging. The description of the synthetic data generation process is also brief, lacking details on the specific parameters and rationale. These presentation issues hinder the reader's understanding of the paper's methodology and findings. In summary, the paper suffers from a combination of methodological and presentation weaknesses that significantly undermine its overall impact. These weaknesses include a lack of clarity, a lack of precise definitions, a lack of methodological detail, a lack of statistical rigor, a limited discussion of limitations, and a reliance on tables for presenting results. These issues need to be addressed to strengthen the paper's contribution to the field. My confidence in these identified weaknesses is high, as they are directly supported by my examination of the paper's content.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should significantly improve the clarity and precision of their writing. This includes providing clear definitions for all key terms and concepts, such as 'need assessment,' 'predictive models,' and 'synthetic dataset.' The introduction should clearly articulate the problem being addressed and the specific research questions being investigated. The methodology section should be reorganized to provide a more logical and coherent narrative flow, clearly explaining each step of the process. The authors should also ensure that all mathematical notation is clearly defined and consistently used throughout the paper. Second, the authors should provide a formal definition of the outcome variables. This definition should specify the exact nature of the outcome being measured, including the units of measurement and the time frame. For example, if the outcome is income, the authors should specify whether it is monthly or annual income, pre-tax or post-tax, and whether it includes other sources of revenue. The authors should also justify their choice of outcome variable and discuss its limitations. Third, the methodology section needs a complete overhaul. The authors should provide a detailed, step-by-step explanation of their experimental design. This should include a clear description of how the synthetic dataset was generated, including the distribution of features and the process for simulating resource needs. The authors should also explain the precise mechanism for random resource allocation, including the probability distribution used and any constraints imposed. For example, if resources are allocated based on a lottery, the authors should specify the number of tickets each individual receives and how the winners are selected. The baseline scenario should be explicitly defined, and the 'Fair needs' intervention should be explained in detail, including the criteria used to determine fairness and the mechanism for allocating resources under this intervention. Including pseudocode or a flowchart would greatly enhance the clarity of the methodology. Fourth, the results section should be revised to include statistical significance tests and confidence intervals. For each finding, the authors should report the p-value and the confidence interval for the effect size. This will allow readers to assess the statistical significance and practical importance of the results. The authors should also provide a more detailed discussion of the limitations of their study. This should include a discussion of potential biases in their synthetic dataset, such as selection bias or measurement error. They should also discuss the generalizability of their findings to other contexts, such as different countries or different public services. For example, if their findings are based on data from Bavaria, they should discuss whether they expect

similar results in other regions of Germany or in other European countries. They should also acknowledge any limitations of their methodology, such as the use of a simplified experimental design or the lack of real-world validation. Fifth, the authors should improve the presentation of their paper by incorporating more visual aids, such as graphs and charts, to illustrate the results and make the paper more accessible to a broader audience. The current presentation relies heavily on tables, which can be difficult to interpret. They should also provide a more detailed explanation of the synthetic data generation process, including the specific parameters used and the rationale behind the choices. The description of the experimental setup should also be more detailed, including the specific evaluation metrics used and the statistical significance of the results. This would enhance the clarity and transparency of the paper and make it easier for readers to understand the methodology and findings. Finally, the authors should consider using a more structured approach to presenting their results, such as using a consistent format for each experiment and clearly stating the research question, methodology, results, and conclusions for each. By addressing these weaknesses, the authors can significantly strengthen their paper and make a more valuable contribution to the field.

? Questions

Based on my analysis, several key questions arise that I believe the authors should address. First, how do you define and measure the outcomes in your study? Are you measuring income, employment status, or a composite measure of well-being? A clear and precise definition of the outcome variable is crucial for interpreting the results and understanding their implications. Second, what is the baseline scenario against which you are comparing your interventions? Is it a scenario where no resources are allocated, or is it a scenario where resources are allocated using a different method? A clear definition of the baseline is essential for understanding the impact of the proposed interventions. Third, how do you ensure that your synthetic data accurately reflects the characteristics and needs of the population you are studying? What steps did you take to validate your synthetic data? The validity of the findings depends on the accuracy and representativeness of the synthetic data. Fourth, what is the precise mechanism for random resource allocation? What probability distribution are you using, and what are the parameters? A detailed explanation of the random allocation process is necessary to understand the experimental design. Fifth, how do you define and measure inequality in your study? Is it the Gini coefficient, the Theil index, or another measure? Please provide a clear definition and justification for your choice. The choice of inequality measure can significantly impact the interpretation of the results. Sixth, what are the specific mechanisms through which predictive models exacerbate inequalities based on migration background, education, and gender? How do you disentangle the effects of these different factors? A deeper analysis of the underlying mechanisms is needed to understand the causes of the observed disparities. Seventh, how do your findings compare to previous research on the use of algorithms in public service provision and their impact on social inequalities? What are the novel contributions of your study? A clear articulation of the paper's contributions is essential for understanding its significance. Eighth, what are the practical implications of your findings for policymakers and practitioners in the public sector? How can they use your research to improve resource distribution and mitigate social inequalities? A discussion of the practical implications is crucial for translating the research into actionable recommendations. Finally, how do you address the potential ethical concerns related to the use of predictive models in need assessment, such as privacy, transparency, and accountability? A thorough discussion of the ethical implications is essential for responsible research in this area. These questions target core methodological choices and seek clarification of critical assumptions, aiming to strengthen the paper's overall contribution.

Scores

Soundness: **1.75**

Presentation: **1.0**

Contribution: **1.75**Rating: **3.0** AI Review from ZGCA will be automatically processed **Summary**

The paper studies how predictive need assessment affects the distribution of public aid within the German Public Employment Service (PES), using 2019 data from Bavaria via the IAB-Experimentation. The authors construct a synthetic dataset centered on whether an individual received a first-round prioritized status (interpreted as privileged access to services) and simulate alternative allocation policies using two baselines: logistic regression and XGBoost (with default settings). They compare observed/prioritized allocation to two policies: needs-based (assign if predicted need ≥ 0.5) and a 'fair chance' variant. The primary outcomes are distributional metrics—variance and Gini coefficients—of income (and to a lesser extent, unemployment and benefit claims), both for the full sample and subgroups (migration background, single parents, young job seekers). The main claim is that predictive allocation, particularly with XGBoost, increases inequality overall and for migrants and single parents, while it benefits young job seekers.

 Strengths

- Addresses an important and underexplored question: the distributional consequences of predictive need assessment in a high-stakes public service.
- Policy-relevant setting with a large, credible data source (IAB-Experimentation) and focus on salient vulnerable groups (migration background, single parents, young job seekers).
- Explicitly examines system-level outcome distributions (variance, Gini) rather than only conventional ML metrics; connects to distributive justice concerns in public resource allocation.
- Clear practical baselines (logistic regression vs XGBoost) that mirror typical PES practice and deployment trajectories.

 Weaknesses

- Unclear and potentially circular labeling: Section 3.2 says the label indicates whether an individual received prioritized access in 2019, but Section 3.2–3.3 then trains a model to predict this label; it is not clear which labels are observed vs predicted, for whom, and how predictions are subsequently used. This ambiguity undermines the empirical strategy.
- Missing causal framework for counterfactual outcomes: Step 2 (randomized allocation and simulation) reassigns resources but does not specify how outcomes (income, benefits, unemployment) are imputed under counterfactual allocations. Without a potential outcomes

model (ATE/CATE, policy value estimation, or uplift modeling), claims about how allocation changes distributional outcomes are not identified.

- Use of default XGBoost without hyperparameter tuning (Section 3.3) compromises fairness of comparisons and internal validity, especially when the core claim is about the impact of adopting predictive models. Literature indicates XGBoost performance and calibration are highly sensitive to tuning.
- Claims of 'significant' effects lack statistical testing (no CIs, hypothesis tests, or uncertainty quantification for Gini/variance changes or subgroup differentials).
- Interventions are under-specified and internally inconsistent: the 'fair chance' policy both restricts to $p \geq 0.5$ and claims 'everyone has at least a 50% chance' (Section 3.4), which cannot simultaneously hold without a clear sampling/budget model. Resource/budget constraints and capacity limits are not formalized.
- Key experimental details are missing: feature set, preprocessing and imputation, train/validation/test protocol, leakage checks, class balance, calibration, subgroup sample sizes, and threshold selection rationale (fixed 0.5).
- Presentation issues reduce clarity: conflicting sample counts (e.g., 275,889 vs 275,929; final sample sizes 85,299 vs 88,522), the misattribution/confusing citation of Athey and Wager (2021) in the data section, and several table entries are difficult to interpret. These issues hinder reproducibility.
- Fairness analysis is narrow (accuracy/precision/recall by subgroup) and disconnected from allocation welfare: no calibration analysis, no evaluation of benefit rates, exposure parity, or policy regret/welfare under budget constraints.

? Questions

- Label construction and usage: Do you observe the 'first-round prioritized status' label for all individuals, or only a subset (e.g., 85,299)? If only a subset is observed, how exactly are labels for the remainder produced (and how is circularity avoided when later predicting need)? Please provide a precise data flow diagram clarifying observed labels vs predicted labels throughout the pipeline.
- Counterfactual outcome modeling: How do you translate reassigning resources (under needs-based or 'fair chance' policies) into counterfactual outcomes for income, benefits, and unemployment? Are you using RCT data to estimate ATE/CATE and then impute potential outcomes? If so, specify the estimators (e.g., doubly robust, T-learner, R-learner) and report policy value estimates with uncertainty.
- Resource/budget constraints: What are the capacity constraints (e.g., fraction treated or service slots) in each policy? How is 'everyone has at least a 50% chance' compatible with restricting to $p \geq 0.5$? Please formalize the policy and budget and report realized treatment rates.
- Model development: What is the full feature set, preprocessing (missingness handling, encoding), and the exact train/validation/test split? How do you guard against leakage? Did you assess calibration (e.g., reliability diagrams, Brier score) for both models and subgroups?
- Hyperparameter tuning: Why not tune XGBoost within cross-validation to reflect responsible deployment? Please provide results with tuned models and conduct sensitivity analyses (e.g., random forest, gradient boosting with tuning, calibrated logistic regression), including calibration and threshold sweep.
- Statistical validation: Please provide confidence intervals and hypothesis tests for all reported 'significant' differences in Gini/variance, including subgroup analyses, and account for multiple comparisons.
- Policy-learning framing: Given access to an RCT, why not use treatment choice frameworks (e.g., Kitagawa & Tetenov 2018; Athey & Wager 2021) or uplift modeling to directly estimate and compare welfare/regret of alternative policies under budget constraints? Can you report policy value (with CIs) for each allocation rule and subgroup?
- Robustness and sensitivity: How sensitive are your inequality conclusions to (i) alternative thresholds (not just 0.5), (ii) alternative inequality metrics (Theil, Atkinson), (iii) different subgroup definitions (e.g., intersectional groups), and (iv) different resource budgets?
- Reproducibility under double-blind: You mention releasing code/data on GitHub. How will you ensure double-blind compliance during review, and what exact artifacts (synthetic data generator, scripts, seeds) will be provided post-acceptance?

Limitations

- Identification gap: Without explicit potential outcomes modeling or policy value estimation, the causal link between reallocation and distributional outcome changes is not established.
- Model configuration risk: Using default XGBoost settings likely underestimates performance and calibration quality and may bias comparisons.
- Unclear intervention budgets: Absent a formal budget/capacity constraint and policy specification, conclusions about real-world distributive effects are hard to interpret.
- Subgroup fairness and calibration: No calibration assessment by subgroup; fairness metrics are limited and not tied to welfare under constraints.
- Statistical uncertainty: No CIs or tests for inequality changes or subgroup differences; multiple comparisons are not addressed.
- External validity: The analysis is limited to one region (Bavaria, 2019) and one institutional context; portability is unclear without replication.
- Potential negative societal impact: Premature conclusions about predictive models deepening inequality, without rigorous identification and sensitivity checks, risk misinforming public debate and policy. Careful causal framing and uncertainty quantification are needed.

Image Evaluation

Cross-Modal Consistency: 20/50

Textual Logical Soundness: 12/30

Visual Aesthetics & Clarity: 6/20

Overall Score: 38/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: Sample size counts are inconsistent across abstract, methods, and conclusion, blocking replication. Evidence: Abstract; Sec. 3.2; Sec. 6 — conflicting sample sizes (85,299 vs 88,522 vs 275,889).
- Major 2: Model/notation mismatch (logistic function with θ estimated by XGBoost) confuses what is being optimized. Evidence: Sec. 3.3 “ $p(x)=\sigma(\theta(x))$... We use XGBoost to estimate θ .”
- Major 3: Metric labels and values in tables are inconsistent (Gini cannot be 99,709.60; placeholder tags appear). Evidence: Table 2 “ineq 99709.60”; entries like “0-0”, “”.
- Minor 1: Undefined acronym. Evidence: Sec. 1 “German public employment service ()”.
- Minor 2: PSHG/PSHG-ID column header is unexplained. Evidence: Tables 1–3 headers “PSHG-Id/PSHG-ID”.
- Minor 3: Misattribution of dataset reference. Evidence: Sec. 3.2 “IAB-Experimentation (Athey and Wager, 2021)”.

2. Text Logic

- Major 1: Contradiction about ground-truth use: label is “received prioritized status,” yet “we do not use actual aid and assistance data.” Evidence: Abstract “label indicates whether a person received a prioritized status”; Sec. 3.4 “We do not use actual aid and assistance data,

but labels”.

- Major 2: Intervention definitions are internally inconsistent (randomized vs deterministic threshold; “same sample” yet “at least 50% chance”). Evidence: Sec. 3.4 Step 2 “randomly assign... Individuals with probability $\geq 50\%$ receive resources... fair chance... same sample... at least a 50% chance”.

- Minor 1: Claims of fairness effects are asserted without consistent, interpretable metrics due to table issues. Evidence: Sec. 4.2 summary vs Table 2 inconsistencies.

3. Figure Quality

- Major 1: Tables are not self-explanatory: duplicated/contradictory headings, missing units, malformed entries, unclear grouping (“Actual needs vs Predicted needs”), making interpretation impossible. Evidence: Table 2 multiple panels with duplicate column names, “ineq” values >1 , stray tags.

- Minor 1: No legend/key for groups or interventions; acronyms and sample sizes per row not defined. Evidence: Tables 1–3 lack legends/captions specifying units and N.

Key strengths:

- Important policy question on distributive impacts of predictive need assessment.
- Ambitious attempt to simulate allocation policies on rich labor market data.
- Focus on vulnerable groups (migration background, single parents, youth) is socially salient.

Key weaknesses:

- Severe cross-modal inconsistencies (sample sizes, model specification, table labeling) prevent verification.
- Intervention design and outcome metrics are ambiguously defined.
- Tables are malformed/illegible for core claims; no confidence intervals or robustness checks.
- Causal interpretation blurred (prediction vs allocation) without a clear identification strategy.

Recommendations (highest impact first):

- Fix sample accounting end-to-end; provide a flow diagram with Ns at each filter.
- Clarify model spec: separate logistic regression from XGBoost; align notation and loss.
- Redesign tables with clear headers, units, valid Gini ranges $[0,1]$, and remove placeholders; add legends.
- Precisely define interventions (deterministic vs randomized), report actual treated shares, and add uncertainty.



Scores

Originality: **3**

Quality: **2**

Clarity: **2**

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 3

 AI Review from SafeReviewer will be automatically processed **Summary**

This paper investigates the impact of using predictive models for need assessment in the German public employment service, specifically focusing on how these models affect the distribution of aid and assistance. The authors utilize a real-world dataset from the IAB-Experimentation in Bavaria, Germany, in 2019, which includes detailed individual and sociodemographic features. They compare two predictive models, logistic regression and XGBoost, against historical need assessments to simulate resource allocation. The core idea is to explore whether these predictive models can improve the targeting of aid and if they might exacerbate existing inequalities. The methodology involves training these models to predict which individuals received active support and assistance, and then using these predictions to simulate resource allocation under different scenarios. The paper's main findings indicate that while predictive models, particularly XGBoost, can improve the overall predictive performance compared to historical methods, they also tend to increase income inequality, especially for vulnerable groups such as individuals with a migration background. The authors highlight that the use of XGBoost, while achieving higher predictive accuracy, leads to a more unequal distribution of resources, suggesting a trade-off between efficiency and equity. The paper's contribution lies in its empirical analysis of a real-world dataset, demonstrating the potential risks of using predictive models in public service allocation. It underscores the importance of considering fairness alongside accuracy when implementing such systems. The study also points to the need for further research into methods that can balance predictive performance with equitable outcomes, particularly for marginalized populations. Overall, the paper provides a valuable case study that raises important questions about the ethical implications of algorithmic decision-making in the public sector.

 Strengths

The paper's primary strength lies in its focus on a highly relevant and under-explored area: the impact of predictive models on resource allocation within the public sector. The authors address a critical question concerning the potential for these models to exacerbate existing inequalities, a topic of significant importance in today's increasingly data-driven world. The use of a real-world dataset from the German public employment service adds considerable value to the study. This dataset, derived from the IAB-Experimentation, provides a rich source of

individual and sociodemographic information, allowing for a more nuanced analysis than might be possible with synthetic or less detailed data. The paper's approach of comparing the performance of predictive models against historical need assessments is also a strength. This comparison provides a practical benchmark, grounding the analysis in the actual practices of the public employment service. Furthermore, the paper's exploration of the trade-offs between predictive performance and equity is a valuable contribution. The finding that XGBoost, while achieving higher accuracy, also leads to increased income inequality highlights the complexities of using predictive models in contexts where fairness is a paramount concern. The paper's focus on vulnerable groups, such as individuals with a migration background, is also commendable. By examining the differential impact of predictive models on these groups, the authors shed light on the potential for algorithmic bias and the need for careful consideration of fairness metrics in model development and deployment. Finally, the paper's overall aim of providing insights into the implications of need assessments for resource distribution is a valuable contribution to the field, raising important questions about the ethical use of AI in public services.

Weaknesses

While the paper presents a valuable case study, several weaknesses significantly impact its overall contribution. Firstly, the paper lacks a clear articulation of its novel contributions beyond the use of a specific dataset. As noted by multiple reviewers, the methodological approach, which involves training classifiers and evaluating outcomes, is not particularly novel. The paper does not adequately explain why this specific case study is important beyond its real-world context. This lack of clear contribution undermines the paper's significance, especially when compared to broader studies that explore resource allocation across multiple domains. Secondly, the paper's methodological description is often unclear and lacks sufficient detail. For example, the process of generating the 'actual needs' labels is not explicitly described, leaving the reader to infer the method. The paper also lacks a clear explanation of the specific features used in the models and the rationale behind their selection. This lack of transparency makes it difficult to assess the validity of the results and hinders reproducibility. The description of the simulation process is also vague, particularly regarding how the different interventions are implemented and how the resource allocation is simulated based on the predicted needs. The paper mentions three interventions but the description is not sufficiently detailed for replication. Furthermore, the paper's evaluation metrics are not always clearly linked to the specific goals of the study. While the paper uses variance and the Gini coefficient to measure income inequality, it does not provide a strong justification for why these metrics are the most appropriate for evaluating the fairness of resource allocation in this specific context. The paper also fails to adequately address the limitations of using income as the sole measure of labor market success. Other factors, such as job stability and job satisfaction, are not considered, which limits the scope of the analysis. The paper also lacks a thorough discussion of the ethical implications of using predictive models in this context. While the paper mentions the potential for increased inequality, it does not delve deeply into the ethical considerations of using these models, such as the potential for bias and the impact on vulnerable populations. The paper also does not provide concrete recommendations for practitioners on how to mitigate the potential negative consequences of using these models. Finally, the paper's presentation could be improved. The writing is sometimes unclear, and the tables are not always well-explained. The paper also lacks a clear narrative that connects the different sections and makes it easy for the reader to follow the authors' line of reasoning. The paper also does not provide sufficient context on the IAB-Experimentation, which is crucial for understanding the data and the experimental setup. The paper assumes familiarity with this dataset, which may not be the case for all readers. Overall, these weaknesses significantly limit the paper's impact and raise concerns about the validity and generalizability of its findings. The lack of methodological detail, the unclear presentation, and the limited discussion of ethical implications all contribute to a paper that falls short of its potential.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should clearly articulate the novel contributions of their work beyond the use of a specific dataset. They should explicitly state what new insights their case study provides and why this specific context is important. This could involve a more detailed discussion of the limitations of existing approaches and how their work addresses these limitations. Secondly, the authors should provide a more detailed and transparent description of their methodology. This

includes explicitly explaining how the 'actual needs' labels were generated, what specific features were used in the models, and the rationale behind their selection. The authors should also provide a step-by-step explanation of the simulation process, including how the different interventions are implemented and how resource allocation is simulated based on the predicted needs. This level of detail is crucial for reproducibility and for the reader to fully understand the experimental setup. Thirdly, the authors should provide a stronger justification for their choice of evaluation metrics. They should explain why variance and the Gini coefficient are the most appropriate metrics for evaluating the fairness of resource allocation in this specific context. They should also consider including other metrics that capture different aspects of fairness, such as equality of opportunity or demographic parity. Furthermore, the authors should broaden their evaluation beyond income as the sole measure of labor market success. They should consider including other factors, such as job stability, job satisfaction, and access to training opportunities. This would provide a more holistic view of the impact of predictive models on labor market outcomes. Fourthly, the authors should include a more thorough discussion of the ethical implications of using predictive models in this context. This should include a discussion of the potential for bias, the impact on vulnerable populations, and the need for transparency and accountability in algorithmic decision-making. The authors should also provide concrete recommendations for practitioners on how to mitigate the potential negative consequences of using these models. This could involve suggesting specific fairness-aware algorithms or proposing methods for monitoring and evaluating the impact of these systems on different demographic groups. Finally, the authors should improve the overall presentation of the paper. This includes ensuring that the writing is clear and concise, that the tables are well-explained, and that there is a clear narrative that connects the different sections of the paper. The authors should also provide more context on the IAB-Experimentation, including its goals, methodology, and key findings, to make the paper more accessible to a broader audience. By addressing these points, the authors can significantly strengthen their paper and make a more meaningful contribution to the field.

? Questions

Several key questions arise from my analysis of this paper. Firstly, what specific criteria were used to define 'actual needs' in the dataset, and how were these criteria operationalized to generate the binary labels? A clearer understanding of this process is crucial for evaluating the validity of the study's findings. Secondly, what specific features were used in the logistic regression and XGBoost models, and what was the rationale behind their selection? Understanding the input variables is essential for assessing the models' predictive power and potential biases. Thirdly, how exactly were the different interventions implemented in the simulation process? A more detailed explanation of the simulation setup is needed to understand how the resource allocation was simulated based on the predicted needs. Fourthly, why were variance and the Gini coefficient chosen as the primary metrics for evaluating income inequality, and what are the limitations of these metrics in this specific context? A more thorough justification of the chosen metrics is needed, along with a discussion of alternative metrics that could provide a more comprehensive view of inequality. Fifthly, what are the specific limitations of using income as the sole measure of labor market success, and what other factors should be considered to provide a more holistic view of the impact of predictive models? Exploring alternative measures of success would provide a more nuanced understanding of the outcomes. Sixthly, what specific steps can be taken to mitigate the potential negative consequences of using predictive models in public service allocation, particularly for vulnerable groups? The paper should provide concrete recommendations for practitioners to address the ethical concerns raised by the study. Finally, what are the limitations of the generalizability of the findings, given that the study is based on a specific dataset from the German public employment service? Understanding the scope of the study's conclusions is crucial for applying its insights to other contexts.



Scores

Soundness: **1.75**

Presentation: 1.75

Contribution: 1.5

Rating: 2.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights